

Assignment No	15
Title	JSP Revision
Objective	1. Use <jsp:include> to display a common header and footer on a JSP page. 2. Display a student's result as Pass/Fail using JSTL <c:if> based on marks. 3. Print numbers from 1 to 10 on a JSP page using JSTL <c:forEach>. 4. Establish a JDBC connection in JSP and check whether the user is valid or not using JDBC. Login application 5. Fetch and display all records from a student table in postgresql and display 6. Write a jsp program for find factorial and check whether given number is prime or not
Roll No	MCA2511

1. Use <jsp:include> to display a common header and footer on a JSP page.

header.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
```

```
<div>
    <h2>Welcome to MET Website Header</h2>
    <hr>
</div>
```

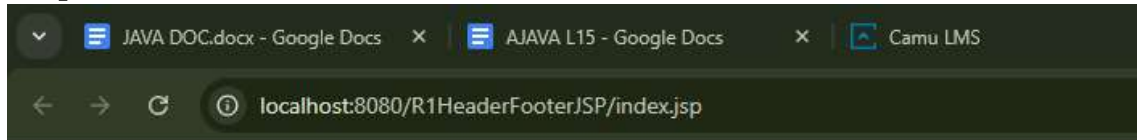
footer.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8"%>
<div>
    <hr>
    <p>Welcome to MET Website Footer</p>
</div>
```

index.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
<!DOCTYPE html>
<html>
<body>
    <jsp:include page="header.jsp" />
    <h3>Welcome to the Home Page</h3>
    <p>This is the main content of the page.</p>
    <jsp:include page="footer.jsp" />
</body>
</html>
```

Output:



Welcome to MET Website Header

Welcome to the Home Page

This is the main content of the page.

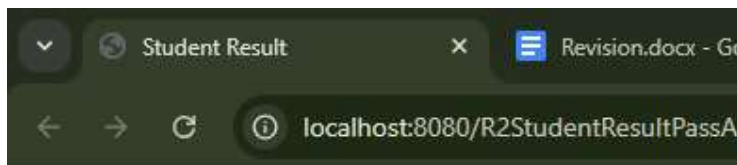
Welcome to MET Website Footer

2. Display a student's result as Pass/Fail using JSTL <c:if> based on marks.

result.jsp

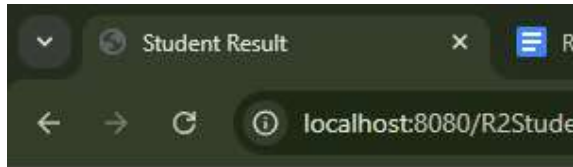
```
<%@ page language="java" contentType="text/html; charset=UTF-8"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>
<!DOCTYPE html>
<html>
<head>
<title>Student Result</title>
</head>
<body>
    <h2>Student Result</h2>
    <form method="post">
        Enter Marks: <input type="number" name="marks" min="0"
max="100"
        required> <input type="submit" value="Check Result">
    </form>
    <c:if test="${not empty param.marks}">
        <h3>Marks Entered: ${param.marks}</h3>
        <c:if test="${param.marks >= 35}">
            <h2 style="color: green;">Result: PASS</h2>
        </c:if>
        <c:if test="${param.marks < 35}">
            <h2 style="color: red;">Result: FAIL</h2>
        </c:if>
    </c:if>
</body>
</html>
```

Output:



Student Result

Enter Marks:

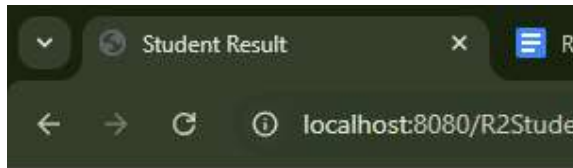


Student Result

Enter Marks:

Marks Entered: 34

Result: FAIL



Student Result

Enter Marks:

Marks Entered: 35

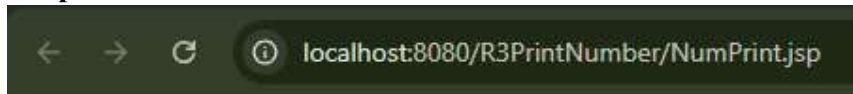
Result: PASS

3. Print numbers from 1 to 10 on a JSP page using JSTL `<c:forEach>`.

NumPrint.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>
<html>
<body>
    <h2>Numbers from 1 to 10</h2>
    <c:forEach var="i" begin="1" end="10">
        <p>${i}</p>
    </c:forEach>
</body>
</html>
```

Output:



Numbers from 1 to 10

1
2
3
4
5
6
7
8
9
10

4. Establish a JDBC connection in JSP and check whether the user is valid or not using JDBC. Login application

login.jsp

```
<!DOCTYPE html>
<html>
<head>
<title>Login Page</title>
</head>
<body>

    <h2>Login</h2>

    <form action="validation.jsp" method="post">
        Username: <input type="text" name="username" required><br>
        Password: <input type="password" name="password"
required><br>
        <br> <input type="submit" value="Login">
    </form>

</body>
</html>
```

validation.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/sql" prefix="sql"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>

<html>
<head>
<title>Login Validation</title>
</head>
<body>

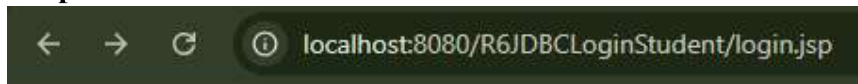
    <sql:setDataSource var="myds" driver="org.postgresql.Driver"
        url="jdbc:postgresql://localhost:5432/postgres?ssl=false"
        user="postgres" password="password" />
```

```
<sql:query var="result" dataSource="${myds}">
SELECT * FROM student WHERE susername = ? AND spassword = ?
<sql:param value="${param.username}" />
<sql:param value="${param.password}" />
</sql:query>
```

```
<c:choose>
  <c:when test="${result.rowCount > 0}">
    <h3 style="color: green;">Login Successful</h3>
    <p>Welcome, ${param.username}!</p>
  </c:when>
  <c:otherwise>
    <h3 style="color: red;">Invalid Username or Password</h3>
    <a href="login.jsp">Try Again</a>
  </c:otherwise>
</c:choose>
```

```
</body>
</html>
```

Output:

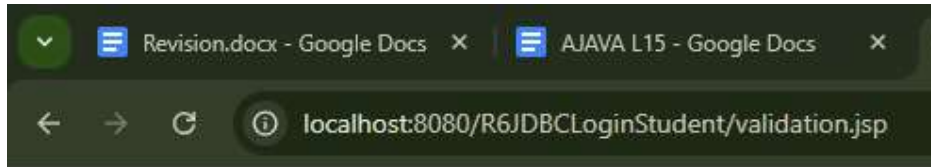


Login

Username:

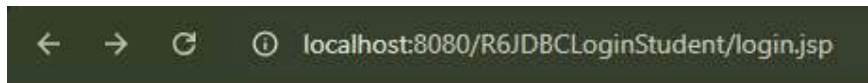
Password:

Login



Login Successful

Welcome, mukund!

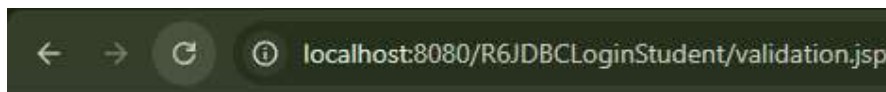


Login

Username:

Password:

Login



Invalid Username or Password

[Try Again](#)

5. Fetch and display all records from a student table in postgresql and display.

index.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/sql" prefix="sql"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>

<html>
<head>
<title>Student Records</title>
<style>
table {
    border-collapse: collapse;
    width: 60%;
}

th, td {
    border: 1px solid black;
    padding: 8px;
    text-align: center;
}

th {
    background-color: #f2f2f2;
}
</style>
</head>
<body>

    <h2>All Student Records</h2>

    <sql:setDataSource var="myds" driver="org.postgresql.Driver"
        url="jdbc:postgresql://localhost:5432/postgres" user="postgres"
        password="password" scope="application" />

    <sql:query var="student" dataSource="{myds}">
        SELECT * FROM student
    </sql:query>

    <table>
        <tr>
            <th>Student ID</th>
            <th>Username</th>
```

```
        <th>Password</th>
        <th>Email</th>
    </tr>
    <c:forEach var="row" items="${student.rows}">
        <tr>
            <td>${row.sid}</td>
            <td>${row.susername}</td>
            <td>${row.spassword}</td>
            <td>${row.semail}</td>
        </tr>
    </c:forEach>
</table>
```

```
</body>
</html>
```

Output:

← → ↻ ⓘ localhost:8080/R5DisplayStudent/index.jsp

Student ID	Username	Password	Email
1	mukund	pass	mg@gmail.com
2	Omkar	13	omkar@gmail.com

6. Write a jsp program for find factorial and check whether given number is prime or not

FactorialPrime.jsp

```
<%@ page language="java"%>
<!DOCTYPE html>
<html>
<head>
<title>Factorial and Prime</title>
</head>
<body>

    <h2>Factorial and Prime Check</h2>

    <form method="post">
        Enter Number: <input type="number" name="num" min="1" required>
        <input type="submit" value="Check">
    </form>

    <%
String n = request.getParameter("num");

if (n != null) {
    int num = Integer.parseInt(n);

    // Factorial calculation
    long fact = 1;
    for (int i = 1; i <= num; i++) {
        fact = fact * i;
    }

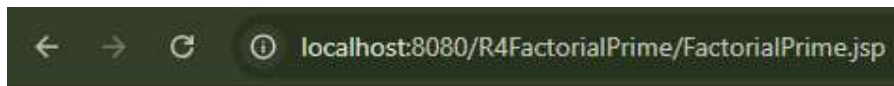
    // Prime check
    boolean isPrime = true;
    if (num <= 1) {
        isPrime = false;
    } else {
        for (int i = 2; i <= num / 2; i++) {
            if (num % i == 0) {
                isPrime = false;
                break;
            }
        }
    }
}

%>
```

```
<h3>
    Factorial of
    <%=num%>
    =
    <%=fact%></h3>

<h3>
    Number
    <%=num%>
    is
    <%=isPrime ? "Prime" : "Not Prime"%>
</h3>
<%
}
%>
</body>
</html>
```

Output:



Factorial and Prime Check

Enter Number:

Factorial of 16 = 20922789888000

Number 16 is Not Prime